

5	(a) (long) straight conductor carrying current of 1 A current/wire normal to magnetic field (for flux density 1 T,) force per unit length is 1 N m^{-1}	M1 M1 A1	[3]
	(b) (i) (originally) downward force on magnet (due to current) by Newton's third law (allow "N3") upward force on wire	B1 M1 A1	[3]
	(ii) $F = BIL$ $2.4 \times 10^{-3} \times 9.8 = B \times 5.6 \times 6.4 \times 10^{-2}$ $B = 0.066 \text{ T}$ (need 2 SF) (<i>g</i> missing scores 0/2, but $g = 10$ leading to 0.067 T scores 1/2)	C1 A1	[2]
	(c) new reading is $2.4\sqrt{2} \text{ g}$ <i>either</i> changes between +3.4 g and –3.4 g <i>or</i> total change is 6.8 g	C1 A1	[2]
6	(a) oil drop charged by friction/beta source	B1	